Data Mining Project
Master in Data Science and Advanced Analytics

NOVA Information Management School
Universidade Nova de Lisboa

Amazing Airlines Customer Analysis

Group 29

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1. Executive Summary

The objective of this report is to analyze AIAI's customer base, to identify distinct segments and design a data-driven loyalty strategy that enhances both profitability and customer satisfaction. Using a combination of behavioural and value-based clustering methods, we identified four key customer segments: Regular Subscribers, Loyal Long-Time Spenders, Enthusiastic New Spenders, and New High Spenders. Each segment shows unique travel and spending patterns, as well as different levels of loyalty, requiring tailored engagement strategies.

Based on these insights, we recommend a three-tiered loyalty program: Current, Crest, and Deep Blue. They are aligned with customers' spending and loyalty profiles. This program is designed to maintain loyalty and increase spending among Regular Subscribers, reward and retain Loyal Long-Time Spenders, build long-term engagement with Enthusiastic New Spenders through targeted incentives, and offer premium experiences and high-value benefits for New High Spenders.

The implementation plan involves a phased rollout, starting with the transition of existing customers, followed by the launch of targeted marketing campaigns. The expected outcome of this strategy is improved per-flight revenue, stronger customer retention, higher engagement, and maximized lifetime customer value.

2. Introduction

In our first report we outlined the need for airlines to offer a competitive and relevant loyalty program, in order to keep up with growing competition and stay relevant. We also showed that the currently implemented loyalty program by AIAI clearly lacks significant separations criteria and fails to distinguish between key customer metrics when it comes to assigning them a loyalty program. Because of this we outlined the goal of finding 3-5 distinct customer clusters, defined by a silhouette score between 50% and 70% as well as a clear interpretability of their business purpose. In the following report we will outline our approach to finding distinguishable customer clusters as well as creating a suitable loyalty program for AIAI.

Feature Engineering

For the data processing we stayed in line with our strategy outlined in the first submission. However, we engineered additional relevant features for Recency (How many months ago the customer flew the last time), Monetary Value (Customer Lifetime Value), Frequency (How often the customer was flying). We created these features to use a RFM (Recency, Frequence, Monetary Value) approach which is commonly used to cluster customer groups in the literature. See Table 3 in the annex for more details about the rationale behind our choices.

3. Methodology

3.1. Feature selection

For our clustering analysis we decided to try two different segmentation approaches, namely a behaviour-based segmentation and a value-based segmentation.

We selected our features based on two main criteria. Firstly, which features fit the given segmentation strategy the best. With bivariate data visualization, we also tried to find patterns in our data. We were looking for pre-existing, well-separated groups of customers. We only found one very clear occurrence of this, and we took advantage of it in the behavioural segmentation, this will be further discussed in paragraph 4.1.

3.2. Outliers

Before applying any algorithms to create clusters, we used DBSCAN to locate the outliers based on the numerical features from our dataset. It was important to deal with outliers, since they can possibly ruin the results of distance-based algorithms. With the optimized DBSCAN model, we found that there were more than 2000 outliers in our data. This was too much data to drop without testing first, so we tried the clustering with and without the outliers. Finally, we decided to drop them, because we got a clearer segmentation of our customer base without them. Without the outliers we got one less group, which made our segmentation more interpretable. The main drawback of it is that we lose some data points that we classified as outliers, however the clustering is most likely going to be a better fit for a large majority of our customers. From the company's perspective, this decision will not lead to lesser profits, since our renewed program will aim to motivate all groups of customers, not just a select few of them.

3.3. Algorithm selection

For each of our segmentation approaches, we tried multiple different clustering algorithms. For both strategies we ended up using k-means clustering, DBSCAN and self organizing maps. We decided on these algorithms, to address grouping patterns which are not circular, or contain groups with different densities. By working with the three clustering methods we are able to cover all these possible data patterns and could still find the optimal clusters. We tried different hyperparameters for each of our algorithms, and compared the best models of all of them using the R^2 metric and the Silhouette Score, as well as visual representations of our clusters.

3.4 Final Cluster Selection

After merging our two main approaches, we run a hierarchical clustering in order to group the similar clusters together and create fewer, more easily interpretable groups. We decided on using hierarchical clustering as it is clearly interpretable and easy to visualize. The reason for this is that after the two different approaches we got some clusters with less than 100 customers in them, which we deemed way too specific for our purpose of creating

meaningful, targetable groups. We selected the final number of clusters using a dendrogram, to see which clusters are closer together.

3.4. Hyperparameter selection

Within each different clustering algorithm, we selected the optimal hyperparameters with the following methods. For K-means clustering, we used the elbow method and the Silhouette Score to get the optimal number of centroids. We also used k-means++ initiation in order to get well-separated clusters. For DBSCAN we used the K-distance method to get an approximate of the range of the best possible eps values, and then we iterated the range to get our final eps value. For self organizing maps we tried different numbers of units and different numbers of clusters.

4. Results & Validation

4.1. Behavioural clustering

For our behavioural segmentation, we wanted to find patterns in how our customers travel. For this, we used the **total number of flights**, the **frequency**, the **mean spent per flight**, the **recency**, the **average flight distance** and the **companion ratio** columns.
After testing the k-means clustering, DBSCAN and self organizing maps, we ended up choosing the density-based approach, DBSCAN. Even though the performance metrics were slightly worse than with our other methods, the DBSCAN was the only one that could capture a visually observable pattern within our data. Our final model had an eps of 0.65 with the minimum number of points set to 12 (twice the number of our clustering columns).

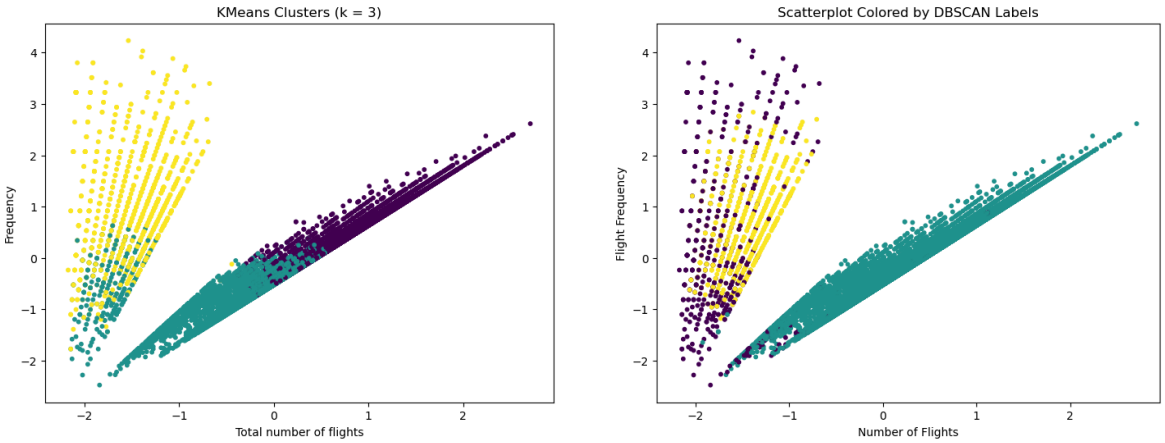


Figure 1: The results of the K-Means and the DBSCAN visualized

For a group of customers it is true that their number of flights is linearly related to their flight frequency. For the other group this is not the case. They generally have a low number of flights during their subscriptions, however their frequency varies from low to very high. As depicted on *Figure 1*, the K-means algorithm could not identify this grouping due to the nature of the algorithm that assumes circular groups of data. Based on the metrics, we decided to go with three final behavioural clusters. The final R^2 value was 17.2%, the final silhouette score was 0.33.

4.2. Value-based clustering

The goal of the Value-based clustering approach is to identify clusters of customers based on their spending habits and the revenue they generate for the airline. For this approach, we decided to use data on the **customer's lifetime value**, **average spent per flight**, and **points redemption rate**. This combination of features allows us to assess how valuable a customer is for the airline. The key things we considered when selecting these features were how much the customer is going to spend, which can be quite well approximated by their lifetime value, as well as the mean spend per flight. We then included the Points Redemption Rate, as it is a key factor in the airline's customer profitability. Well, spending customers who collect a lot of points but don't spend them (low Points Redemption Rate) effectively save the Airline money, as they are not consuming offerings that are free to them. All these factors could be key to finding the right customers for our improved loyalty program. According to our metrics, the self organizing maps and the k-means produced similar results, there was no single 'best' clustering by both our metrics. We ended up using k-means for its simplicity since the results were similar compared to the SOM. For the final model we used k-means++ with 15 initiations.We ended up with two final clusters. The R^2 value for our final value-based clustering was 38.8%. The silhouette score was 0.52.

4.3. Combined clustering

After the separate clustering, we combined the approaches together, resulting in 6 clusters. Some of these had less than a hundred customers in them, making the clustering imbalanced. After trying hierarchical clustering on our 6 clusters, we ended up merging two pairs of clusters, resulting in our final four clusters. This way our grouping became a bit more balanced and more interpretable. The final R^2 score was 0.21 with the silhouette score being 0.23. We can see a visualization of the differences between our clusters in *Figure 2*.

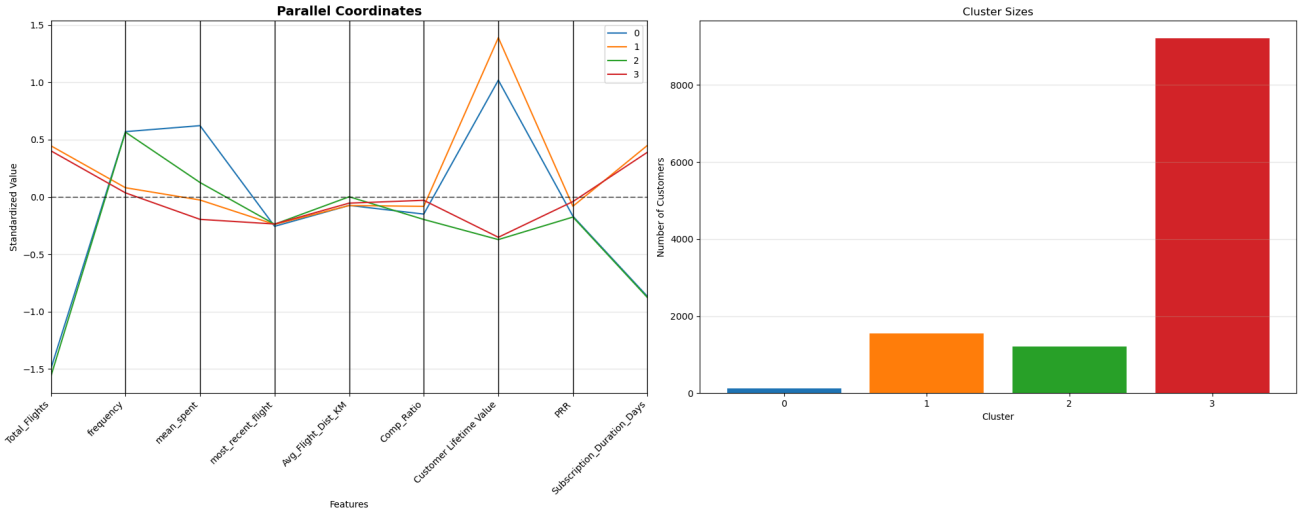


Figure 2: Visualization of the differences in the feature values and the cluster sizes between our final clusters

The most important variables for separating our data were the number of flights, the frequency, the mean spent per flight, the customers lifetime value and the length of the subscriptions. The means of these features were the most different among our four groups.

4.4. The final clusters

Out of our final clusters, the most populous group is the 'Mom & Pop Fliers', our regular customers. In general they have been our subscribers for a longer than average period of time, however their value has not been high from the company's standpoint.

Our next group is the 'Loyal, long time spenders'. They are our long time customers with a high lifetime value. They tend to spend more than our 'Mom & Pop Fliers' and over time, this results in a much higher lifetime value.

The 'Enthusiastic new spenders' are the active new customers, who fly often. Since they are new subscribers, their lifetime values are relatively low, however, they show a promising tendency.

Finally, our 'New high spenders' are just like the previous group, the difference is that they pay more per flight on average, resulting in a high lifetime value, even though they are also relatively new customers.

We make the observation that our first two and second two groups are very similar to each other, mainly their lifetime values differ. In section five, we give strategic recommendations for each group.

4.5. Fuzzy clustering

Fuzzy clustering consists of creating clusters in which each observation belongs to every cluster with a certain degree of membership, defined by a membership function. This approach is particularly useful in settings where data points are densely packed, making crisp separations problematic: small differences in the feature space may lead to large changes in hard assignments (for example, a point with 51% vs. 99% proximity to a cluster would be treated identically in a hard clustering framework).

In this project, we applied the **Fuzzy C-Means (FCM)** algorithm with two main objectives:
(i) to compare the fuzzy labels with those obtained through previous clustering, and
(ii) to assess whether the resulting cluster structures differ significantly once uncertainty is explicitly taken into account.

The optimal number of clusters was determined by maximizing the **Fuzzy Partition Coefficient (FPC)**, which measures the degree of separation between clusters, and minimizing the **Partition Entropy (PE)**, which captures the uncertainty of the partition. The algorithm was applied separately to the same feature sets used in the two perspectives:

- **Behavioural**: frequency, mean spent, most recent flight, average flight distance (km), and companion ratio;
- **Value-based**: customer lifetime value, mean spent, and point redemption rate.

Inspection of the evaluation metrics (see Annex F1 and F2) showed that, for both perspectives, the optimal solution consists of **two clusters**, consistently with the final crisp clustering solution. The fuzzy framework, however, also provides access to the **membership matrix**, allowing us to analyze the degree of uncertainty associated with each customer's assignment.

Membership values were used to investigate two aspects: whether fuzzy labels were aligned with our previous labels, and whether a substantial share of customers exhibited ambiguous assignments. Following a common rule of thumb, membership values between **0.33 and 0.67** were flagged as *mixed*, indicating that the observation could not be clearly assigned to a single cluster.

The distribution of mixed memberships shows that approximately **one third of behavioural observations** were classified as mixed, while only **about one sixth of value-based observations** exhibited similar ambiguity. This suggests that customer behaviours are highly overlapping and concentrated in dense regions of the feature space, whereas value-related characteristics display stronger separation.

To compare the two fuzzy solutions, we combined behavioural and value sub-clusters and aligned them with the previous labels, using the hierarchical dendrogram reported in the Annex at F3. Labels were assigned based on the dominant membership (greater than 0.5). The resulting comparison revealed notable differences: for value-based clustering, **1,159 observations** received a different label compared to k-means, while for behavioural clustering this number increased to **6,043**.

Within the value-based segmentation, **796 observations** were classified as mixed (membership below 0.67), whereas in the behavioural case only **1,194 out of 6,043** differing observations were mixed. The two perspectives were then combined to obtain a final segmentation with **four clusters**.

Comparing the final fuzzy-based labels with the previous solution showed that **6,011 observations** were assigned to a different cluster. An agreement-versus-difference plot (see Annex F4) was used to identify which clusters diverged the most, and a cross-tabulation matrix (Annex F5) further illustrated how observations moved between clusters.

The magnitude of these differences suggests that the two approaches are not directly comparable. Instead, the fuzzy solution can be interpreted as an alternative segmentation. Analysis of the descriptive statistics (see Annex F6) shows that the fuzzy-based clusters retain a structure similar to the crisp segmentation, with the main difference being the emergence of two very large profiles (families and business-oriented customers). This outcome can be explained by the different objective function fuzzy c-means, with it explicitly allowing for overlap between groups.

A substantial proportion of customers exhibits mixed membership, particularly along behavioural dimensions. Specifically, **34.31%** of observations show mixed behavioural

assignments, compared to **20.76%** for value-based assignments (Annex F7). This asymmetry indicates that behavioural characteristics are inherently more ambiguous and overlapping, while value-related variables such as spending and CLV are more stable and better separated.

At the cluster level, value-based clusters 0 and 2 display low levels of mixed membership, suggesting homogeneous and well-defined financial profiles. In contrast, clusters 1 and 3 show higher ambiguity, reflecting heterogeneous value trajectories and less coherent economic profiles. Behavioural mixing, instead, is high and relatively uniform across all clusters, confirming that behavioural patterns form a continuum rather than discrete segments.

Overall, value-based clustering yields cleaner and more interpretable segments, whereas behavioural ambiguity appears to be a global property of the data rather than cluster-specific. These results support the use of fuzzy clustering to capture uncertainty and overlap, especially for behavioural features, and highlight that cluster-level averages alone may be insufficient to describe certain groups. For practical applications such as targeting or causal analysis, value-based labels should therefore be considered more reliable.

From an operational perspective, it is advisable to start with **hard clustering** to define clear and actionable customer groups for marketing initiatives. Fuzzy clustering can then be employed as a second-level analysis to refine strategies, identify overlapping behaviours, and design personalized promotions based on probabilistic memberships, ultimately improving engagement and customer value.

5. Strategic Recommendations

Creating an Airline loyalty program is a tricky task, requiring a balance between customer demands, high loyalty, and not exceeding the business's costs. Aspects of point collection options across the different programs can offer a massive opportunity to gain a competitive advantage and separate an airline from its competitors. That is why we start our recommendations by analysing the existing customer base and its attributes. The existing customer base of Amazing Airlines can be separated into four distinct types of airline customers. The key metrics for separating them from a business perspective are the Spending Level (how much they spent) and their Attachment to the Airline (loyalty). Using these two measurements, we can visualize them on a two-by-two matrix. This visualization should be used to provide an overview of the "as is" situation and to easily understand how our final strategy aligns with AIAI's current customers. Additionally, the matrix represents customer clusters based on their Airline Revenue relevance, with the best customers located at the top right, showing High levels of spending while maintaining high Airline loyalty.

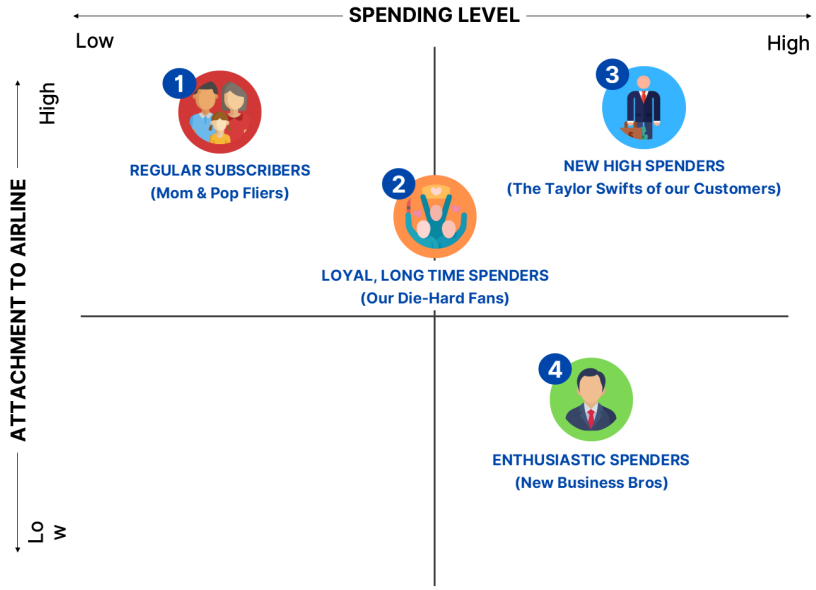


Figure 3: Airline Customer Matrix

Starting at the top left of the Matrix, we have the customer group, which is very loyal to our Airline but doesn't tend to spend much on travel. We mapped this behaviour to a typical family on holiday. Very similar to this group is another cluster further to the right; this group of clients tends to be less loyal but also spends a bit more. This sort of behaviour could also apply to similar customer attributes as our first loyal fans, just with a bit more consumption and maybe price sensitivity.

Our top clients are located in the top right of the matrix; they are very loyal to our company and spend high amounts on travel. This type of pattern is an excellent type of behaviour for loyalty programmes and, if implemented correctly, can guarantee the airline a stable source of income.

Finally, we have our fourth customer cluster on the bottom-right side of the matrix. These are customers who have a high average spending but a low loyalty to our Airline. Considering this behaviour, these are most likely business travellers who book flights long in advance and factor in costs, but then spend money on services during their travels.

Loyalty program recommendations

Given our customers' preferences and behaviour, we recommend a three-pillar loyalty program. We took the liberty of developing a naming convention inspired by water and nature. The loyalty program structure starts with Current < Crest < Deep Blue, with our most valuable customers in the Deep Blue group.

Loyalty Program "Current":

Starting with the first group, the "Current" customers. In this segment, we align the needs of customer cluster one and parts of cluster two. The goal for this program is to maintain loyalty and increase their spending. For this, we provide special points offers for "comfort" purchases made for their travels. The advertising strategy for these customers should align with this and offer special seating upgrades, as well as food or lounge access.

Loyalty Program "Crest":
Crest customers are unsure about our offerings, but when they travel, they spend a lot. This type of pattern is promising, as it suggests the potential for high profitability if we can build a strong relationship with them. For this purpose, the Crest program strongly focuses on building loyalty by offering special point-collection options for Bundle Flight purchases and Flights booked far in advance. The marketing approach supports this offering by promoting flights based on customers' purchasing patterns and bundle deal discounts. For this customer group, we accept lower flight profitability and expect to earn their money from additional purchases and increased loyalty.

Loyalty Program "Deep Blue":
The Deep Blue program is our De-lux offering for our best and most loyal customers. Here, we don't directly promote new services or upgrades; instead, we focus on reasons for them to fly somewhere. Special offerings for concerts or other events will make them feel appreciated by the Airline and provide an excellent reason for them to travel. Based on their previous behaviour, they will spend a high amount on their airline travel, ensuring overall profitability. For points rates, we recommend a higher-than-usual points offering compared to other programs, making this program also attractive for people not dedicated to the "jetset" life.

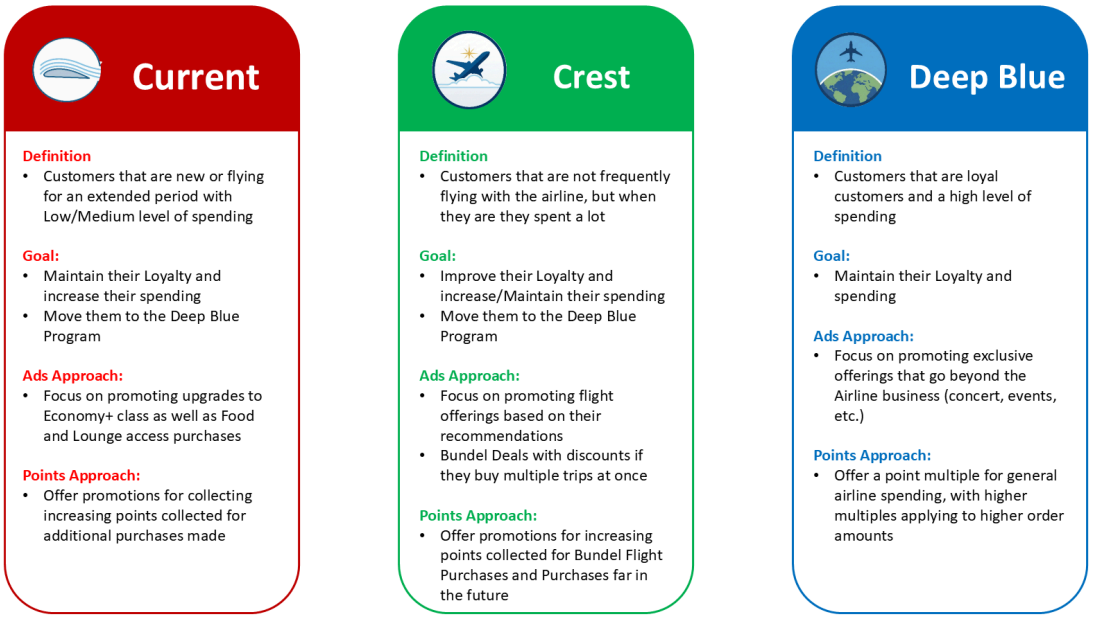


Figure 4: Loyalty Program Groups and their Features

Implementation recommendations

Given the complexity of our recommendations, we suggest a step-by-step rollout of the new loyalty program. In the first phase, AIAI should begin transitioning existing customers to the new loyalty program and automatically assign all new customers to it. In phase two, we prepare the marketing campaigns for the new groups, which we will launch in phase three. By using this provided approach, AIAI can ensure that its existing customer base and operations

are not disrupted by the change, while also guaranteeing a timely implementation of the measures.

6. Conclusion

Our analysis demonstrates that Amazing International Airlines (AIAI) can significantly enhance its loyalty program by adopting a data - driven segmentation approach. From separate behavioural and value-based clustering, followed by a combined hierarchical methodology, we identified four distinct customer clusters: “Regular subscribers”, “Loyal, Long-Time Spenders”, “Enthusiastic New Spenders”, and “New High Spenders”. These clusters are defined by key metrics such as flight frequency, customer lifetime value, average spend per flight, and subscription length, providing a clear and useful structure in order to tailor loyalty offerings.

The application of fuzzy clustering highlights even more the underlying ambiguity in behavioural patterns, indicating that a subset of customers manifest overlapping behaviours. Value - based segmentation, in contrast, showed more stable and interpretable clusters, suggesting that financial aspects provide a more reliable foundation for loyalty program targeting. Integrating fuzzy clustering with hard clustering can help as a second-level improvement, allowing personalized offers and probabilistic targeting for customers with ambiguous behaviours.

From a business perspective, these insights guided the development of our three tier program: “Current”, “Crest”, “Deep Blue”. They are designed to strengthen customer attachment and optimize revenue. Each tier addresses specific customer needs and behaviours, balancing loyalty incentives with profitability objectives.

Despite these promising results, several limitations must be acknowledged. Our analysis relied on historical transaction data, which may not fully capture changing customer behaviours or external market influences. The clustering algorithms, while robust, are sensitive to feature selection and hyperparameter choices, and future behaviour may diverge from past patterns. Moreover, behavioural ambiguity suggests that some customer preferences may not be fully captured in static clusters.

Future research could address these limitations by incorporating dynamic data sources, such as real-time booking patterns, seasonal effects, or feedback from customer surveys. Moreover, exploring a focus on integrating clustering with predictive modeling to improve segmentation accuracy. For example, combining fuzzy clustering results with models that predict future customer behaviour, could help AIAI anticipate how customers shift between segments over time. This would make the loyalty strategy more data-driven and adaptable to changing customer patterns.

In conclusion, our segmentation based framework offers AIAI a clear route to optimize loyalty program design, arranging customer behaviours with business strategy while highlighting areas for improvement and research.

Appendix A

Table: 1 Summary statistics for raw CustomerDB

Column Name	Meaning	Missing values
Loyalty#	Unique identifier for members	-
First Name	Customer's first name	-
Last Name	Customer's last name	-
Customer Name	Customer's full name	-
Country	Customer's country of residence	-
Province or State	Customer's province or state of residence	-
City	Customer's city of residence	-
Latitude	Geographic latitude of the city	-
Longitude	Geographic longitude of the city	-
Postal code	Customer ZIP code	-
Gender	Customer's gender	-
Education	Customer's education level	-
Location Code	Customer's residence classification	-
Income	Customer's annual income	20
Marital Status	Customer's marital status	-
Loyalty Status	Current loyalty program	-
Enrollment Date Opening	Joining date of the program	-
Cancellation Date	Date when the customer left the program	14327
Customer Lifetime Value	Monetary value of the customer	20
Enrollment Type	How they found the program	-

Table 2: Summary statistics for raw FlightsDB

Column Name	Meaning	Missing values
Loyalty#	Unique identifier for members	-
Year	Year of the recorded flight activity	-

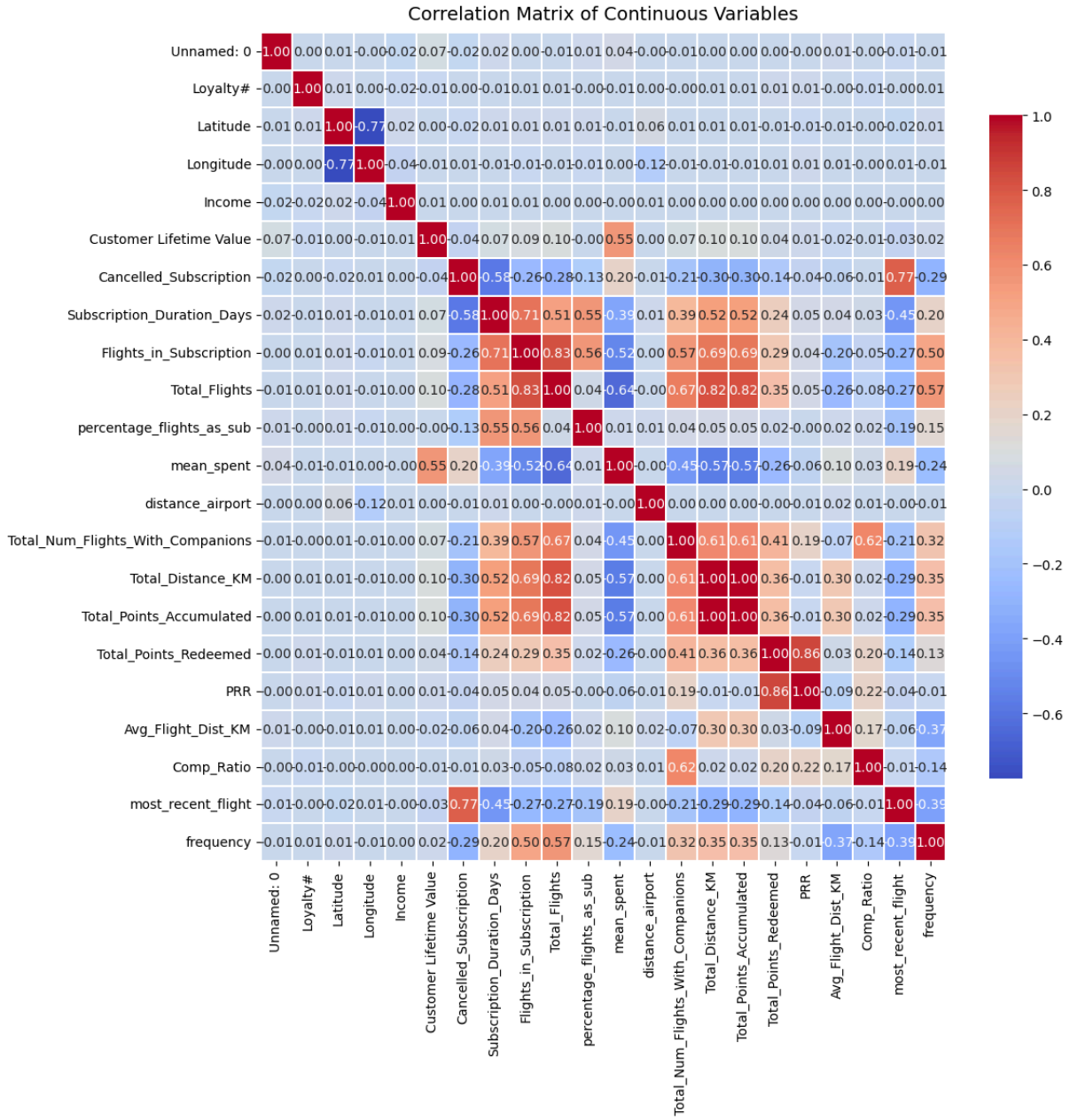
Month	Month of the recorded flight activity	-
Year Month Date	First day of the month with recorded activity	-
Flights	Number of flights taken in the month by the customer	-
Flights with companions	Number of flights taken in the month by the customer with companions	-
Distance KM	Distance travelled in the mont	-
Points accumulated	Loyalty points earned by the customer in the month	-
Points redeemed	Loyalty points redeemed in the mont	-
Dollar cost points redeemed	Dollar value of the points redeemed in the month	-

Table 3: Summary table for engineered features

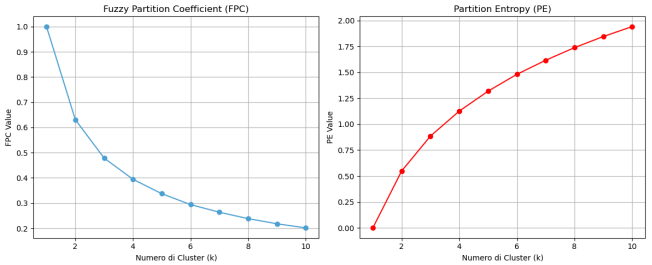
Column	Meaning
Subscription duration	Obtained by flagging the customers that cancelled their subscription and finding the duration of it by subtracting the two dates
Flight in subscription period	We took the number of flights for each customer that happened inside the period of subscription
Percentage of flights in the subscription	We calculated the ratio of flights inside the subscription period to identify possible changes in the behaviour
Companion flights ratio	We calculated the percentage of flights with a companion. We can identify people that travel a lot with someone else, like a family member
Average money spent for flight	How much a customer spends on average for a flight, we can identify spending patterns and this could propably help us to understand flight patterns or value based insights.
Total points earned	Cumulative sum of the points earned by the customer. We can use this feature alone or to create the point's redemption rate
Total points redeemed	Cumulative sum of the points redeemed by the customer. Same as above
Point's redemption rate	Percentage of points redeemed by the customer. Useful for understanding how much a customer participates in our programs

Distance from airport	How far the customer lives from the airport (rural customers). Mainly a demographic feature that can tell us if long distances from airports can change the flight behaviour
Average travel distance	Average flight distance for every customer. We can try to understand the flight patterns of our customers.
Recency	Obtained by calculating how much time has passed since the last flight of our customer. Useful in RFM optic
Frequency	How much the customer flew on average after the first flight. Useful in RFM optic

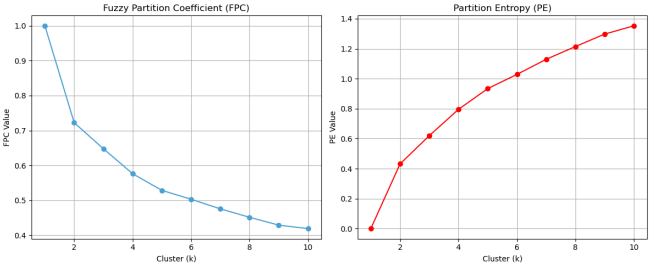
Figure 1 Correlation matrix



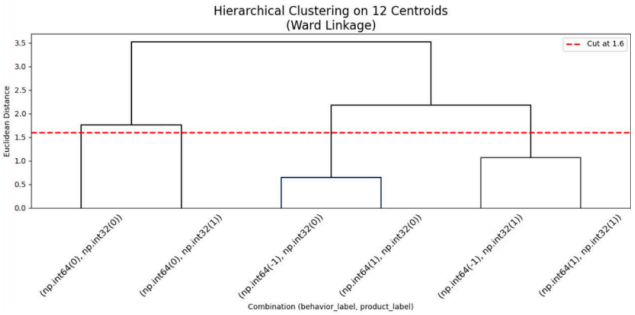
Fuzzy clustering graphs



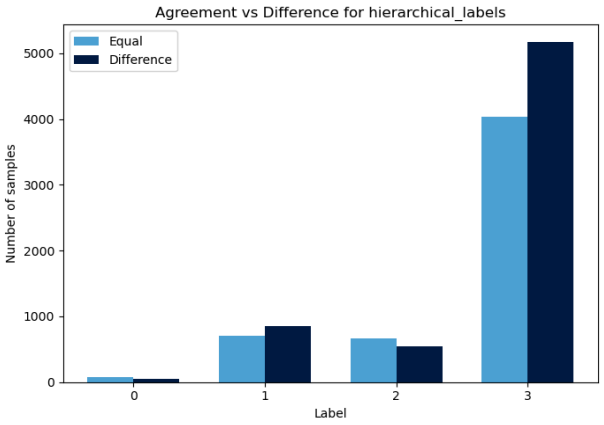
Graph F1: behavioural fuzzy clustering measures



Graph F2: value fuzzy clustering measures



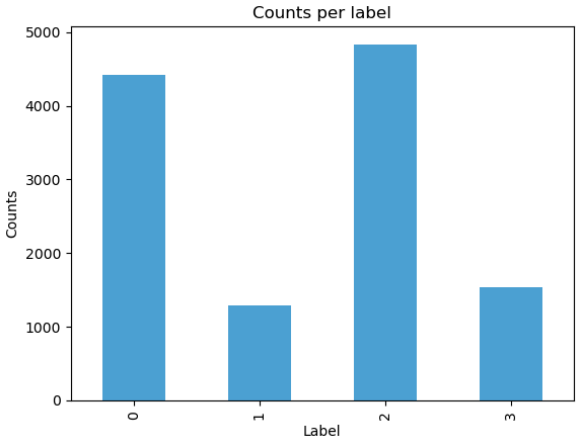
Graph F3: hierarchical clustering transformation



Graph F4: differences of labels for hierarchical and fuzzy clustering

fuzzy_cluster_label	0	1	2	3
hierarchical_labels				
0	80	48	0	0
1	847	700	0	0
2	86	65	667	391
3	526	482	4166	4031

Graph F5: crosstab of labels for hierarchical and fuzzy clustering



Graph F6: labels of fuzzy clustering

```
Total observations with mixed behavioural marker: 34.31%
Total observations with mixed value marker: 20.76%
fuzzy_cluster_label
0    18.204432
1    32.200772
2    17.008070
3    30.279402
Name: mixed_value, dtype: float64
fuzzy_cluster_label
0    36.431479
1    36.525097
2    32.795365
3    31.124107
Name: mixed_behavioural, dtype: float64
```

Graph F7: statistics of our fuzzy labels

ANNEXES

Annex AI Usage Statement

Claude: Brainstorm visualization ideas and example syntax generation

Report Proofreading (Grammarly for grammar checks)

ChatGPT: Troubleshooting the code and helping with visualization;

All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

Annex Contribution Statement

Moritz Baldauf (Student ID: 20250370)

- Creation of EDA Dashboard
- Strategy Recommendations

Libero Biagi (Student ID: 20250349)

- Feature understanding and cleaning
- Partial feature engineering
- Data analysis section in the report
- Fuzzy Clustering part

Jenny Cubelo, (Student ID: 20250431)

- Help with Cluster analysis
- Creation of Video
- Executive summary and Conclusion in the report

Marcel Hajdú, (Student ID: 20250377)

- Helping with data cleaning
- Feature engineering
- Visualization of pre-processed data
- Results paragraph in the report

All members contributed to collaborative discussions and ideation.

Annex Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission

Annex Interactive EDA Dashboard

The goal of this Interactive dashboard is to provide a management overview of our cluster results and to enable easy interpretation of the insights. The dashboard includes a 3D scatter plot of customer segments, a radar plot of customer profiles, and Boxplots of the clusters' distributions of selected values. In addition, the dashboard allows filtering for demographic attributes such as gender, education, marital status, and income. As well as selecting multiple clusters and axes for the visualization. It is also possible to use the scaled or unscaled features. When management uses the filter, they are also provided with key summary statistics for the subcluster they created (number of customers, average income, average lifetime value, average spend), as well as an option to download the data they are viewing.

3D-Scatter Plot

Key visualization of the clusters in a three-dimensional space. It can show two different clusters at once, with options for the user to change all three axes and scale or unscale them. The dashboard is interactive and can be moved. For each datapoint, a detailed pop-up appears when the user hovers over it.

Radar Plot

Shows the difference between our key features in the two clusters displayed in the Scatter plot. The key attributes are: Total travel Distance, Income, Customer Lifetime value, mean spend, and frequency. The goal of this plot is to get a quick overview of how the clusters differ in customer behaviour and attributes.

Exploration Controls

The Filter segmentation sidebar is the control center of the dashboard; any selection or removal of filters updates all the existing elements instantly. It allows filtering the dataset, changing which clusters are shown, and adjusting the scatter plot's axes and scaling them.

Attributed Deep Dive

This is a simple visualization of a single cluster attribute at a time. The goal is to allow users to more deeply explore the clusters and gather insights into relevant features and differences between the clusters.

Customer Details Inspection

It is a panel that provides customer information when a user selects a datapoint in the scatterplot. In addition to information on the city, income, gender, education, and marital status, it provides a plot of the fuzzy cluster assignments for this point. This information can be used to assess how certain our results are that the customer belongs to a single cluster.

Instructions on how to run the dashboard

1. Install dash in your environment
2. After installing the required libraries, the dashboard can be started by running the dedicated cell.
    - However, since this window might be too small for the user preference, it is also possible to run the dashboard in a dedicated .py file. This execution will lead to the dashboard being started in a browser which can be accessed with the address: http://127.0.0.1:8050